

TEST BANK

CHAPTER 2

PLANKTON AND PLANKTON COMMUNITIES

Matching

In this section, you are to select from the choices listed in column B the term or phrase that best matches the term or phrase listed in column A and then to place the letter for the selected choice from column B in the space provided to the left of the corresponding item in column A.

[Note to instructors: Here, the choices are matched and thus should be randomly assorted before the exam is prepared.]

Column A		Column B
___ 1. dinoflagellates	_____→	A. zooxanthellae
___ 2. diatom	_____→	B. domoic acid
___ 3. gelatinous zooplankton	_____→	C. marine snow
___ 4. nauplius	_____→	D. larval copepod
___ 5. osmosis	_____→	E. water moves across a membrane
___ 6. radiolarians	_____→	F. geodesic dome
___ 7. <i>k</i>	_____→	G. extinction coefficient
___ 8. collective limiting factors	_____→	H. hydrography
___ 9. DOC, bacteria, and nanozooplankton	_____→	I. microbial loop
___ 10. nutrient rich	_____→	J. eutrophic

Multiple Choice

Circle the choice that best completes or answers the given questions.

[N.B.: The correct answer to each of the questions is indicated by an asterisk. For most of the questions, the letter "a" is the correct choice; reorder the choices as you see fit.]

1. Which of the following phytoplankton construct an external covering or shell composed of two halves of silicon dioxide?

 - a. Diatoms *
 - b. Dinoflagellates
 - c. Cyanobacteria
 - d. Radiolaria
 - e. Marine snow
2. Which of the following colors of light reaches the greatest depth in the ocean before being completely absorbed?

 - a. blue *
 - b. red
 - c. green
 - d. violet
 - e. yellow

108 Test Bank for Marine Biology

3. Which of the following phytoplankton are responsible for the condition known as red tides?
 - a. Dinoflagellates *
 - b. Diatoms
 - c. Cyanobacteria
 - d. Coccolithophores
 - e. Marine snow
4. Which of the following phytoplankton possess bioluminescent species?
 - a. Dinoflagellates *
 - b. Diatoms
 - c. Cyanobacteria
 - d. Coccolithophores
 - e. Marine snow
5. Radiolarians are most closely related to which of the following organisms?
 - a. Amoeba *
 - b. Malaria
 - c. Dinoflagellates
 - d. Diatoms
 - e. Coccolithophores
6. Foraminiferans belong to which of the following groups?
 - a. Protozoa *
 - b. Ctenophora
 - c. Cnidaria (Coelenterata)
 - d. Arthropoda
7. Critical depth is that depth at which total gross photosynthesis of phytoplankton in the water column is _____.
 - a. = total respiration *
 - b. > total respiration
 - c. < total respiration
8. The greatest number of planktonic organisms is found in which one of the following phyla?
 - a. Arthropoda *
 - b. Mollusca
 - c. Cnidaria
 - d. Ctenophora
 - e. Chordata
9. The classes Thaliacea and Larvacea are planktonic, gelatinous filter feeders of phylum _____.
 - a. Chordata *
 - b. Cnidaria
 - c. Ctenophora
 - d. Mollusca
 - e. Arthropoda
10. Langmuir convection cells are produced only when wind speeds are in excess of _____.
 - a. 3 m/sec *
 - b. 5 m/sec
 - c. 15 m/sec
 - d. 30 m/sec
11. The amount of _____ in the world's oceans constitutes one of the largest reserves of organic carbon on our planet.
 - a. DOC *
 - b. marine snow
 - c. copepods
 - d. chordates
12. When the net 1° productivity is greater than zero, then which of the following statements is correct?
 - a. Gross 1° productivity = Respiration
 - b. Gross 1° productivity > Respiration *
 - c. Gross 1° productivity < Respiration
 - d. None of the above
13. The compensation depth is that depth in the water column where _____.
 - a. rate of photosynthesis = rate of respiration *
 - b. rate of photosynthesis > rate of respiration
 - c. rate of photosynthesis < rate of respiration
 - d. None of the above
14. The dominant copepod of the North Atlantic Ocean is _____.
 - a. *Calanus finmarchicus* *
 - b. *Neocalanus plumchrus*
 - c. *Neocalanus cristatus*
 - d. None of the above
15. _____ is the term used to describe all the factors relating to water that affect phytoplankton production.
 - a. Hydrography *
 - b. Winkler method
 - c. Compensation intensity
 - d. Convection cells
 - e. Marine snow
16. In waters that are thermally stratified, the upper water layer is _____ than the lower layers.
 - a. less dense *
 - b. more dense
 - c. cooler

17. ____ is an important proximal stimulus that initiates and controls diel vertical migration of zooplankton.
 - a. Light *
 - b. Temperature
 - c. Food
 - d. The presence of predators
 - e. None of the above
18. Which one of the following types of organisms is considered to be the dominant group of the zooplankton throughout the world's oceans?
 - a. Copepods *
 - b. Mollusks
 - c. Squids
 - d. Worms
 - e. Fish
19. In seawater, the compensation depth for net community photosynthesis is the point at which the rate of photosynthesis _____.
 - a. = rate of respiration *
 - b. > rate of respiration
 - c. < rate of respiration
20. In an ecosystem, as energy is passed from one trophic level (food level) to another, the amount of usable energy _____.
 - a. increases
 - b. decreases *
 - c. remains the same
21. Red tides are caused by which one of the following?
 - a. Dinoflagellates *
 - b. Bacteria
 - c. Diatoms
 - d. Alabama's football team
 - e. Copepods
22. Consumption of bacteria by heterotrophic nanoflagellates has been termed (the) _____.
 - a. microbial loop *
 - b. chloroplast symbiosis
 - c. remote sensing
 - d. biological conditioning
 - e. seasonal succession
23. Patches of plankton can vary from a few meters to ____ in size.
 - a. dozens of meters
 - b. hundreds of meters
 - c. kilometers
 - d. hundreds of kilometers *
24. ____ are masses of rotating water that have broken off from a larger ocean current and have moved into another water mass.
 - a. Rings *
 - b. Coastal fronts
 - c. Patches
 - d. Microbial loops
 - e. Marine snow
25. Vertical turbulence in the water column caused by frictional resistance generated when water flows over the shallow continental shelf is called _____.
 - a. coastal front *
 - b. advection
 - c. eddies
 - d. rings
 - e. Langmuir cells
26. The zone of rapid temperature decline with depth is known as (the) _____.
 - a. thermocline *
 - b. rings
 - c. Ekman spiral
 - d. hydrography
 - e. Langmuir cells
27. In copepods, there are ____ stages of nauplii followed by ____ stages of copepodids.
 - a. 6, 5 *
 - b. 5, 6
 - c. 5, 5
 - d. 10, hundreds of
 - e. 2, 2
28. Which of the following types of animals has members that are predators?
 - a. Radiolarians *
 - b. Ctenophores
 - c. Jellyfish
 - d. Chaetognath
 - e. All of the above

110 Test Bank for Marine Biology

29. On an areal basis, the net primary production of the ocean is only about ____ that of the terrestrial environment.
- 25%*
 - 10%
 - 75%
 - 50%
30. The ____ are probably the most numerically abundant phytoplankters in the open oceans.
- prochlorophytes *
 - coccolithophores
 - zooxanthellae
 - dinoflagellates
 - diatoms

True/False

Mark each of the given statements with the letter *T* if it is true and *F* if it is false.

[Instructors: In some cases a few options are included for the same question.]

- [True]* The Chaetognatha, or arrowworms, are voracious predators of copepods. *[False]* jellyfish
- [False]* The Chaetognatha, or arrowworms, are herbivorous zooplankton.
- [False]* Marine snow is unimportant in the open ocean.
- [True]* Copepods can detect the presence of an algal cell without touching it.
- [True]* The compensation depth is approximately the depth in the water column to which 1% of the incidental light penetrates.
- [False]* The compensation depth is approximately the depth in the water column to which 10% of the incidental light penetrates.
- [True]* The dark bottles in the two light–dark bottle techniques (oxygen and ^{14}C) do not measure or correct for the same thing.
- [False]* The dark bottles in the two light–dark bottle techniques (oxygen and ^{14}C) measure or correct for the same thing.
- [True]* Different species of phytoplankton have different optimal light intensities for maximal photosynthesis.
- [True]* Viruses are composed of nucleic acids (either DNA or RNA) and protein, but little to no lipid.
- [False]* All species of phytoplankton have the same optimal light intensities for maximal photosynthesis.
- [True]* Rings are masses of rotating water that have broken off from a larger ocean current and have moved into another water mass.
- [True]* The auxospore is a structure produced by diatoms that functions to reestablish the original size of the species.
- [False]* The auxospore is a structure produced by copepods that functions to reestablish the original size of the species.
- [True]* Marine snow is composed of gelatinous material lost by the gelatinous zooplankton.
- [False]* Marine snow is restricted to the colder portions of the Earth.
- [True]* The standing crop is the total amount of an organism's biomass per unit volume of water at a given time.
- [True]* Osmosis is the passage of water across a semipermeable membrane.
- [True]* Certain squids are capable of excluding from their body fluids heavy ions that are less dense than seawater.
- [False]* The Portuguese man-of-war is an animal that is capable of excluding from their body fluids heavy ions that are less dense than seawater.
- [True]* Certain Ctenophores have a flat shape, which acts as a kind of passive flotation mechanism.
- [False]* Diatoms reproduce by continuous binary fission. *[False]* continuous sexual reproduction

23. [False] Moving from the equator to the south, the main planktonic biotic provinces encountered in the Pacific Ocean are, in order, the equatorial zone, transition zone, subantarctic zone, and South Pacific central gyre zone.

Short Answer (Fill in the Blank)

1. Red tides are produced by some types of dinoflagellates.
2. All copepods have a similar life cycle composed of the following four distinct phases or stages: egg, naupliar stages, copepodid stages, and adult.
3. Dinoflagellates are sometimes responsible for causing the mass mortality of certain types of inshore marine life by their production of powerful neurotoxins.
4. While most phytoplankton are eukaryotes, one group, the cyanobacteria, consists of prokaryotes.
5. The standing crop is the total amount of an organism's biomass in a given volume of water at a given time.
6. Net production is the total amount of PS-fixed carbon left after losses from plant respiration.
7. Locomotion in ctenophores is accomplished by rows of large cilia called ctenes.
8. Wind blowing over the surface of a body of water at speeds above 3 m/sec produces Langmuir convection cells.
9. Hydrography is a concept that comprises all those factors that act to move water masses around the oceans.
10. The Winkler method may be used to monitor oxygen concentration when using the light–dark bottle oxygen technique to measure primary productivity.
11. The prochlorophytes are probably the most numerically abundant phytoplankters in the open oceans.
12. Along the region inshore to the Gulf Stream, rings rotate clockwise, while on the Sargasso Sea side, they rotate counterclockwise.
13. The ammonium ion is often used by planktonic organisms to replace heavier ions, thereby increasing their buoyancy.

Essay Discussion Questions

Develop complete answers to these questions.

1. How can the pore size of the plankton net influence our understanding of the relative importance of planktonic organisms in the open ocean?
2. How did the advent of sonar influence our understanding of oceanic processes?
3. Discuss the relative significance of eukaryotic and prokaryotic phytoplankton.
4. What is the difference between plankton and nekton? How does size play a role in the difference between organisms that fit into one or the other of these groups?
5. What determines the maximum and minimum sizes of individuals of a particular diatom species?
6. What is an auxospore? Discuss the significance of auxospores to organisms possessing them.
7. What function(s) do spines serve in diatoms?
8. Oceanic phytoplankton populations tend to occur in patches. How are these patches generated?
9. How might silicon be a limiting factor to diatoms?
10. Discuss the taxonomic problems with the taxon known as Kingdom Protocista (Protista). How does chloroplast symbiosis further complicate this question?
11. How are red tides responsible for massive mortality of fish and invertebrates? How does this mortality occur? That is, what are the biological processes responsible for the death?
12. In what ways are people affected by red tides?

112 Test Bank for Marine Biology

13. Certain nanoflagellates are not capable of photosynthesis. How do they survive, and of what importance are they to oceanic food webs?
14. Are the blue-green algae really algae?
15. Besides aiding in slowing their sinking rate, what function(s) do the very large antennae serve in the copepods?
16. What is a copepod? To what common freshwater, marine, and terrestrial organisms are they related? To which phylum do they belong?
17. Why are copepods important in marine systems?
18. Describe the general life cycle and developmental stages of copepods. As part of your essay, present a well-labeled figure that diagrams the life cycle of a copepod.
19. Compare and contrast the two models of copepod feeding, discussing where the older research was in error and why that work would have naturally concluded that the filtering currents created by copepods would reprocess water. [*Instructors: Naturally, this question requires that you offer some explanation of the older model.*]
20. To which phylum do the organisms that create foraminiferan (for example, *Globigerina*) and radiolarian ooze belong, and how do they feed?
21. How do radiolarians resemble the invention of Buckminster Fuller (referred to as buckminsterfullerene, buckyballs, and a geodesic dome)?
22. Why, on an areal basis, is the net primary production of the oceans much less than that of the terrestrial environment? Discuss.
23. Why are planktonic organisms often distributed in patches? Discuss.
24. How does chloroplast symbiosis change our view of protozoans?
25. Comb jellies possess eight radially arranged, longitudinal rows of locomotory cilia called ctenes. Describe how these structures can be worked by the animal so that it can rotate in any direction.
26. Explore the concept of hydrography as it relates to marine productivity.
27. Name two different flotation mechanisms employed by animals that involve the metabolic production of a specific material. Discuss.
28. Compare and contrast the flotation mechanisms of spine production and the production of buoyant materials. Discuss.
29. Lipids can act as both food reserves and aids in buoying organisms. How might the storage of fat and oil droplets be detrimental to copepods? [*Hint: Think of light shining on a drop of oil surrounded by a medium possessing a different index of refraction (such as water).*]
30. Describe a typical seasonal cycle of primary productivity in the upper waters of the ocean.
31. How does grazing increase nutrient concentration in the upper water mass of oceans? Discuss.
32. What is biological conditioning, and how does it influence plankton communities?
33. How does the compensation depth vary on a 24-hour basis? How does the compensation depth vary on a seasonal basis?
34. Distinguish between the compensation depth and the critical depth, and explain their significance to phytoplankton.
35. Compare the light/dark bottle oxygen method of measuring primary production with the ^{14}C method.
36. In the two light and dark bottle techniques, how do the dark bottles differ? What do the dark bottles in the oxygen method and in the ^{14}C method correct (measure)?
37. Is temperature an important factor in determining the amount of primary production (photosynthesis) in marine environments? Discuss.
38. How did flow cytometry technology influence our understanding of the nanoplankton?
39. Why is the sky blue? Why are the oceans sometimes blue and sometimes green? Why is the sky red in the morning and at night?

40. Why can you get sunburned on a cloudy day? How is it possible for one to get a severe sunburn on a snow field? Of what significance are these concepts to organisms that live on or close to the surface of the water? [*Instructors may wish to restructure this question to explore the relationship of the ozone layer and UV light. Appropriate references that you may wish to have your students consult are listed as follows:*

McKenzie, R., Connor, B., and Bodeker, G. 1999. Increased summertime UV radiation in New Zealand in response to ozone loss. *Science* 285: 1709–1711.

Smith, R.C. et al. 1992. Ozone depletion: ultraviolet radiation and phytoplankton biology in Antarctic waters. *Science* 255: 952–959.]

41. What colors can you see at a depth of about 30 meters without a flashlight? With a flashlight? Is there a difference? Why?
42. Discuss the ecological importance of the picoplankton.
43. Why does water with substantial input of glacial flour (rock ground up by glacial action) appear an aquamarine (blue-green) color?
44. Speculate as to why the chlorophyll maximum occurs at greater depths (100–150 m) in the tropics than elsewhere?
45. Why are the oceans considered by some to be like deserts? Is this a realistic view? Discuss.
46. How does the thermocline contribute to the establishment of the seasonal pattern of production in tropical seas? (*in temperate seas?*)
47. Discuss the development of the spring bloom.
48. What are the differences between the rings formed from the Gulf Stream on the oceanic side and the rings formed on the continental side?
49. Why is the population density, or standing crop, of phytoplankton a poor or misleading measure of primary production?
50. Compare and contrast biomass, standing crops, and primary production ratios in marine and terrestrial systems.
51. What is vertical migration, and what controls it? Discuss.
52. Why do zooplankton undergo a diel vertical migration? (*Review the hypotheses that explain diel vertical migration in zooplankton.*)
53. Why did it take divers looking at the plankton in the natural environment of the open ocean to discover marine snow?
54. Summarize the ideas first reviewed by McLaren and then by Haney that attempt to explain why zooplankton undertake nocturnal migration.
55. Based on what factors does the light-damage avoidance hypothesis seek to explain diel vertical migration of zooplankton? In avoiding damaging light, how are the zooplankton subjecting themselves to increased risk of predation?
56. Discuss Hardy's explanation for the evolution of vertical migration.
57. Discuss the energetic-advantage hypothesis of Enright (and others) as an explanation of the evolution of vertical migration.
58. Why are there never climax communities in marine planktonic communities, as there are in terrestrial systems?
59. Discuss three arguments that attempt to explain succession of phytoplankton.
60. Plot on an x – y coordinate axis a figure that illustrates the seasonal succession of phytoplankton. (Be sure to label the axes.) In essay form, explain the factors that influence the rise and fall of phytoplanktonic populations.
61. Why does phytoplankton production often fall off in the summer after the spring bloom?
62. Describe the environmental conditions that foster the fall pulse of phytoplankton production.

114 Test Bank for Marine Biology

63. What is a red tide, and what types of organisms cause it? What types of toxins are produced by red-tide organisms, and what do these toxins do to animals that consume them?
64. How does a red tide affect coastal shellfish industry?
65. Certain soil bacteria cause the diseases gas gangrene, botulism, and tetanus. Similarly, the toxins produced by the red-tide organisms can do serious damage to the nervous systems of large vertebrates. Speculate as to why these microbes produce compounds that are so highly toxic to vertebrates despite the fact that the vertebrates are rarely, if ever, a direct threat to the microbes.
66. What vitamins would you think to be most abundant in the surface waters of the open ocean? Speculate as to what vitamins would be least abundant in the deep abyssal or hadal waters. [*Hint: What types of vitamins are produced by plants?*]
67. Are open-water marine ecosystems controlled by top-down or bottom-up processes?
68. Discuss how satellite oceanography has helped our understanding of the world's oceans.
69. What is the significance of DOC in the world's oceans?
70. Describe the production of a thermocline, and explain why a thermally stratified water column will resist the mixing action of the wind.
71. Discuss why inshore waters tend to have higher levels of 1° production than offshore waters.
72. Summarize three effects that zooplankton populations have on phytoplankton populations.
73. How is it possible that, while the biomass of phytoplankton is declining due to intense grazing by zooplankton, the rate of primary production is increasing? Discuss.
74. How can a change in the surface of resistance of a body slow its sinking?
75. Compare and contrast decomposition in terrestrial and marine ecosystems.
76. Calculate the relative percent saving in body weight if a planktonic animal replaces SO_4^{2-} with Cl^- . [*Instructors may wish to provide a periodic table, if one is not available in the classroom.*]
77. What are two problems commonly encountered with the light–dark bottle oxygen technique?
78. What are two problems commonly encountered with the ^{14}C technique?
79. Describe the importance of the microbial loop to the carbon cycle in plankton food webs.
80. If a population of 10 dinoflagellates begins to grow and goes through five generations (cell divisions or doublings), how many dinoflagellates will be present at the end of the growth period?
81. Speculate as to why the major planktonic biomes in the Pacific Ocean are associated with the major gyre systems.
82. What causes patchiness of plankton in the open ocean?
83. Describe the significance of the term e to measuring net primary productivity.
84. How are foraminiferans and radiolarians important to deep-sea marine sediments?
85. The ammonia ion is often used by planktonic organisms to replace other ions. What function does this replacement serve?
86. Challenge Question: Why is net primary production of the oceans only about 25% of that of the terrestrial environment?
87. Challenge Question: Why is it more difficult to come to a full understanding of the life histories of meroplankton than of holoplankton?
88. Challenge Question: Discuss how viruses are important to the microbial loop.
89. Challenge Questions: What is marine snow? Is there a freshwater counterpart to marine snow? Is there a terrestrial counterpart to marine snow? Discuss.
90. Challenge Question: What problem(s) would be encountered if primary productivity were measured using the ^{14}C method and the pH of the water was less than 5.5?